

Activity #14

US Armed Forces Data Exploration

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1 Exploring the Armed Forces

We will use the wrangled data to explore whether sex and rank are independent of one another through a two-way frequency table.

Figure 1: Figure 2. Two-way frequency table of Army enlisted personnel by sex and rank.

Table 1: Two-way frequency table of Army Enlisted personnel by Sex and Rank

Rank	Female	Male	Total
Corporal OR Specialist	15143	79234	94377
First Sergeant OR Master Sergeant	1472	9482	10954
Private	5662	29767	35429
Private First Class	10229	43775	54004
Sergeant	10954	54803	65757
Sergeant First Class	4410	30264	34674
Sergeant Major OR Command Sergeant Major	394	2865	3259
Staff Sergeant	7363	49502	56865
Total	55627	299692	355319

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1.0.1 Narrative Interpretation

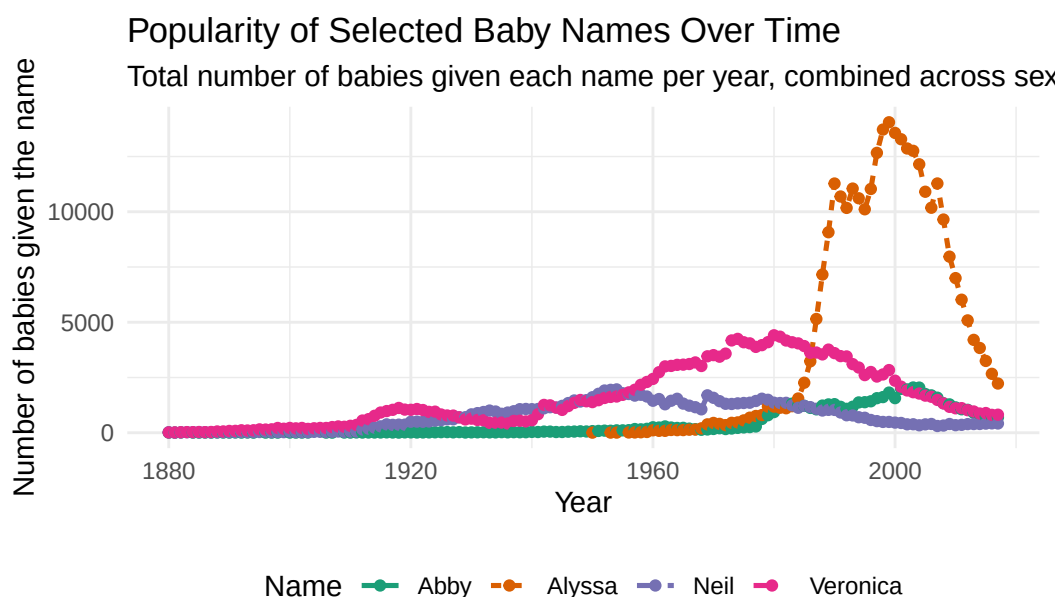
The two-way frequency table comparing sex and rank among Enlisted soldiers in the U.S. Army shows how representation varies across pay grades. Each row corresponds to a specific enlisted rank (such as Private, Corporal, or Staff Sergeant), and the columns report the number of male and female soldiers in each role, along with row and column totals. Across the table, the overall pattern reveals that male and female counts increase and decrease together across ranks rather than diverging sharply. Ranks with larger numbers of males, such as the junior enlisted levels, also tend to have larger numbers of females relative to other ranks, while senior enlisted positions show smaller and fairly similar counts for both sexes.

This pattern suggests that, within this subgroup, sex and rank appear to be largely independent. There is no single rank that is disproportionately dominated by one sex relative to the overall distribution; instead, the male-to-female ratio is reasonably consistent from the lower enlisted grades up through the senior enlisted grades. While males remain more numerous overall in the Army, the relative distribution of males and females across the enlisted ranks does not show strong evidence of rank-specific concentration by sex, which is consistent with the idea that promotion through the enlisted ranks occurs similarly for male and female soldiers in this data set.

2 Popularity of Baby Names

To explore baby name popularity in a meaningful way, we chose the first names of our instructional team — Veronica, Alyssa, Abby, and Neil — so the analysis stays directly connected to our course context. We look at the overall popularity of each name across time without separating by sex.

Figure 2: Figure 2. Time series of the popularity of the names Veronica, Alyssa, Abby, and Neil in the United States.



Source: babynames package (U.S. Social Security Administration data)

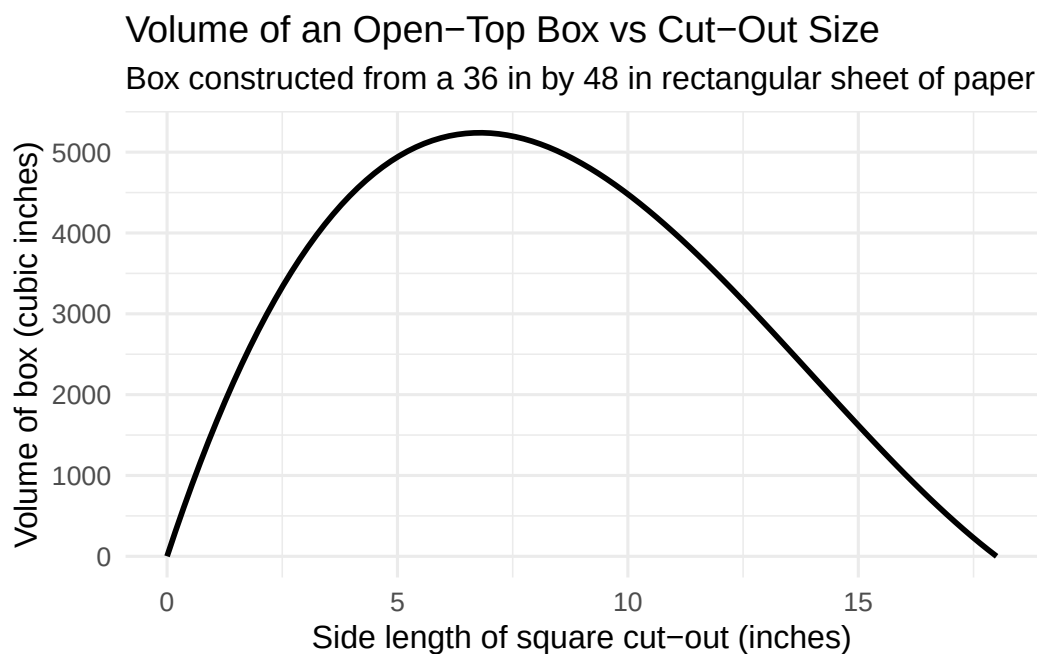
2.0.1 Narrative: Popularity of Names

Figure 2 shows how the popularity of the names Veronica, Alyssa, Abby, and Neil has changed in the United States over time. I chose these names because they match the first names of the instructional team for this course, which makes the visualization personally relevant and easier to connect to the class context. The plot displays the total number of babies given each name per year, and the different line styles and colors make it easy to compare the four trends.

From the visualization, we can see that Alyssa experiences a sharp rise in popularity during the 1990s before declining quickly, while Veronica shows a steady mid-century peak and gradual decrease. Abby grows more moderately, and Neil remains a consistently uncommon name throughout the years. These patterns help us understand how naming trends emerge and fade across generations. The visualization clearly illustrates these changes and supports comparisons across the four names.

3 The Box Problem

Figure 3: Figure 3. Volume of an open-top box as a function of the cut-out size for a 36 in by 48 in sheet of paper.



#Narrative for the Box Problem

Figure 3 displays the volume of an open-top box as a function of the cut-out size when starting with a 36 inch by 48 inch sheet of paper. The curve rises from zero, reaches a single clear maximum, and then decreases again as the cut-out becomes too large, which visually shows that only moderate cut-out sizes produce useful box volumes. From the peak of the graph, the maximum volume occurs at a cut-out size of approximately 6 inches, which gives a maximum box volume of roughly 5200 cubic inches. This visualization makes it easy to see both where the maximum occurs and how

quickly the volume declines when the cut-out is either too small or too large, helping us understand the trade-off between base size and height for the box.

4 Self Reflection

Honestly, I have learned a lot in this course. I started to think about all the R concepts that I learned, starting with simple ones like functions with the box problem, which seemed so familiar. However, then I moved into harder concepts like data cleaning and wrangling with the army and family data using tidyverse, which really showed me how helpful it is to be structured in what I do. I also learned a lot about creating data visualizations with the babynames with ggplot2, and that really made things a lot more complicated to do, but easier to see, and that was really rewarding. The greatest thing I learned from this class is debugging, however, especially with all the problems that I had to overcome with my code; but I learned how to do so, and I believe I benefited quite a lot from it.

5 Code Appendix

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#####

# Code Appendix: Armed Forces - Two-way Frequency Table

#####

# Load packages

library(google sheets4) # For loading Armed Forces Data
library(rvest)          # For scraping Armed Forces Ranks
library(tidyverse)      # For data wrangling and visualizing
library(janitor)        # For making two-way table
library(knitr)          # For creating professional tables
library(kableExtra)     # For styling professional tables

# Use "public / no login" mode for reading the sheet

gs4_deauth()

sheet_url <- "[https://docs.google.com/spreadsheets/d/19xQnI1cBh6Jkw7eP8YQuuicM1VDF7Gr-nXCb5qbv"

# Grab the header rows (title + actual headers)

forces_hdr_raw <- read_sheet(
  ss          = sheet_url,
  col_names  = FALSE,
  n_max      = 3
```

```

)

# Grab the core data block under the headers

forces_raw <- read_sheet(
  ss      = sheet_url,
  col_names = FALSE,
  skip    = 3,
  n_max   = 28,
  na      = "N/A*"
)

# Construct useful column names and clean the forces table

branches_vec <- rep(
  x      = c("Army", "Navy", "Marine Corps", "Air Force", "Space Force", "Total"),
  each = 3
)

forces_colnames <- paste(
  c("", branches_vec),
  forces_hdr_raw[3, ],
  sep = "."
)

names(forces_raw) <- forces_colnames

forces_tidy_base <- forces_raw %>%
  dplyr::rename(PayGrade = `.Pay Grade`) %>%
  dplyr::select(!dplyr::contains("Total")) %>%
  dplyr::filter(
    PayGrade != "Total Enlisted",
    PayGrade != "Total Warrant Officers",
    PayGrade != "Total Officers",
    PayGrade != "Total"
  ) %>%
  tidyr::pivot_longer(
    cols      = !PayGrade,
    names_to  = "BranchSex",
    values_to = "Count"
  ) %>%
  tidyr::separate_wider_delim(
    cols  = BranchSex,
    delim = ".",
    names = c("Branch", "Sex")
  )

```

```

# Pull and reshape the rank lookup table

rank_page <- read_html("https://neilhatfield.github.io/Stat184_PayGradeRanks.html")

rank_tables <- rank_page %>%
  html_elements("table") %>%
  html_table()

ranks_raw <- rank_tables[[1]]

ranks_raw[1, 1] <- "MetaType"

rank_header <- ranks_raw[1, ]
names(ranks_raw) <- rank_header[1, ]

ranks_core <- ranks_raw[-c(1, 26), ]

ranks_tidy <- ranks_core %>%
  dplyr::select(!MetaType) %>%
  tidyr::pivot_longer(
    cols      = !`Pay Grade`,
    names_to  = "Branch",
    values_to = "Rank"
  ) %>%
  dplyr::mutate(
    Rank = na_if(Rank, "--")
  )

# Join the forces data with rank titles

forces_with_rank <- dplyr::left_join(
  x = forces_tidy_base,
  y = ranks_tidy,
  by = dplyr::join_by(PayGrade == `Pay Grade`, Branch == Branch)
)

# Build group-level and individual-level data sets

forces_groups <- forces_with_rank

forces_individuals <- forces_with_rank %>%
  dplyr::filter(!is.na(Count)) %>%
  tidyr::uncount(weights = Count)

# Two-way frequency table for Army enlisted (E ranks)

```

```

#| label: fig-armed-forces-freq
#| fig-cap: "Figure 2. Two-way frequency table of Army enlisted personnel by sex and rank."
#| fig-alt: "A contingency table showing counts of Army enlisted service members by rank (rows)

army_enlisted_ind <- forces_individuals %>%
filter(
  Branch == "Army",
  grepl("^E", Pay.Grade)
)

army_rank_sex_tab <- army_enlisted_ind %>%
  tabyl(Rank, Sex) %>%
  adorn_totals("row") %>%
  adorn_totals("col")

army_rank_sex_tab %>%
  kable(caption = "Two-way frequency table of Army Enlisted personnel by Sex and Rank") %>%
  kable_styling(full_width = FALSE)

#####

# Code Appendix: Popular Baby Names

#####

#| label: fig-popular-baby-names
#| fig-cap: "Figure 3. Time series of the popularity of the names Veronica, Alyssa, Abby, and Neil"
#| fig-alt: "Line chart showing the number of babies named Veronica, Alyssa, Abby, or Neil in t

library(babynames)
library(dplyr)
library(ggplot2)

selected_names <- c("Veronica", "Alyssa", "Abby", "Neil")

baby_names_clean <- babynames %>%
  filter(
    name %in% selected_names
  ) %>%
  group_by(year, name) %>%
  summarise(
    total_births = sum(n),
    .groups      = "drop"
  )

baby_names_plot <- ggplot(
  data      = baby_names_clean,

```

```

mapping = aes(
  x      = year,
  y      = total_births,
  color  = name,
  linetype = name
)
) +
geom_line(linewidth = 0.9) +
geom_point(size = 1.5) +
scale_color_brewer(
  palette = "Dark2",
  name     = "Name"
) +
scale_linetype_discrete(
  name = "Name"
) +
labs(
  title      = "Popularity of Selected Baby Names Over Time",
  subtitle   = "Total number of babies given each name per year, combined across sex",
  x          = "Year",
  y          = "Number of babies given the name",
  caption    = "Source: babynames package (U.S. Social Security Administration data)"
) +
theme_minimal(base_size = 11) +
theme(
  legend.position = "bottom"
)

baby_names_plot

#####

# Code Appendix: Box Problem - Volume Function Plot

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#| label: fig-box-volume
#| fig-cap: "Figure 4. Volume of an open-top box as a function of the cut-out size for a 36 in
#| fig-alt: "Smooth curve showing the volume of an open-top box on the vertical axis versus the

library(ggplot2)

getBoxVolume <- function(x, L = 48, W = 36) {
  volume <- x * (L - 2 * x) * (W - 2 * x)
  return(volume)
}

```



```

box_domain <- data.frame(
  x = c(0, 18)
)

box_volume_plot <- ggplot(
  data      = box_domain,
  mapping   = aes(x = x)
) +
  stat_function(
    fun      = getBoxVolume,
    args     = list(L = 48, W = 36),
    linewidth = 1
  ) +
  scale_x_continuous(
    limits = c(0, 18)
  ) +
  labs(
    title     = "Volume of an Open-Top Box vs Cut-Out Size",
    subtitle  = "Box constructed from a 36 in by 48 in rectangular sheet of paper",
    x         = "Side length of square cut-out (inches)",
    y         = "Volume of box (cubic inches)"
  ) +
  theme_minimal(base_size = 12)

box_volume_plot

```